

Perceptron Rule Example

- ❑ Assume a 3 input perceptron plus bias (it outputs 1 if $\text{net} > 0$, else 0)
- ❑ Assume a learning rate c of 1 and initial weights all 0: $\Delta w_i = c(t - z) x_i$
- ❑ Training set
 - 0 0 1 $\rightarrow$ 0
 - 1 1 1 $\rightarrow$ 1
 - 1 0 1 $\rightarrow$ 1
 - 0 1 1 $\rightarrow$ 0

Pattern	Target	Weight Vector	Net	Output	ΔW
0 0 1	1 0	0 0 0 0			

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1 1 1	1	0 0 0	0	0	1 1 1
1 0 1	1	1 1 1	3	1	0 0 0
0 1 1	0	1 1 1			

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1 0 1	1	1 1 1	3	1	0 0 0
0 1 1	0	1 1 1	3	1	0 -1 -1
0 0 1	0	1 0 0			

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Pattern	Target	Weight Vector	Net	Output	ΔW	
0 0 1 1	0	0 0 0 0	0	0	0 0 0 0	1st epoch
1 1 1 1	1	0 0 0 0	0	0	1 1 1 1	
1 0 1 1	1	1 1 1 1	3	1	0 0 0 0	
0 1 1 1	0	1 1 1 1	3	1	0 -1 -1 -1	
0 0 1 1	0	1 0 0 0	0	0	0 0 0 0	2nd epoch
1 1 1 1	1	1 0 0 0	1	1	0 0 0 0	
1 0 1 1	1	1 0 0 0	1	1	0 0 0 0	
0 1 1 1	0	1 0 0 0	0	0	0 0 0 0	